

Ocean Wind Field Estimation

from Sentinel-1 SAR Imagery Over Indian Coasts

Study Areas: Tamil Nadu Coast | Gujarat Coast

Dataset: Copernicus Sentinel-1 IW GRD · Wind Model: CMOD5.n GMF

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1. Introduction

Offshore wind energy is among the fastest-growing renewable sectors in India. Accurate characterisation of the marine wind resource along the country's coastlines is critical for optimal siting of offshore wind farms. Traditional in-situ measurements are sparse and expensive; reanalyses such as ERA5 have coarse spatial resolution (~25 km). Sentinel-1 SAR imagery fills this gap with kilometre-scale snapshots of ocean surface wind fields at no operational cost.

This project delivers a production-grade REST API that accepts a user-specified date and Area of Interest (AOI), retrieves Sentinel-1 SAR imagery through Google Earth Engine, applies the CMOD5.n GMF to retrieve wind speed, and returns geo-referenced wind vectors with a publication-quality visualisation.

1.1 Problem Statement

Ocean wind field estimation using SAR imagery over Indian coastal areas for wind farm planning. Study areas: Tamil Nadu and Gujarat coasts. Deliverable: an API providing wind field vectors for a user-supplied date and area of interest.

1.2 Objectives

- Acquire Sentinel-1 IW GRD (VV polarisation) imagery via Google Earth Engine for the selected AOI and date window.
- Retrieve ocean surface wind speed by inverting the CMOD5.n Geophysical Model Function.
- Estimate wind direction through SAR-image gradient streak analysis.
- Expose the complete pipeline as a FastAPI REST endpoint.
- Validate retrieved wind speeds against ERA5 reanalysis reference data and generate a validation table.
- Generate publication-quality wind vector field maps.

2. Dataset

2.1 Sentinel-1 SAR

Sentinel-1 is a C-band (5.405 GHz) SAR constellation operating at a centre wavelength of 5.6 cm. The IW GRD product provides 10 m spatial resolution VV/VH backscatter over a 250 km swath. Incidence angles range from 29° to 46°. Data are freely available through the Copernicus Open Access Hub and indexed in Google Earth Engine (COPERNICUS/S1_GRD).

2.2 ERA5 Reanalysis (Validation Reference)

ERA5 provides hourly 10 m neutral-stability wind speed at $0.25^\circ \times 0.25^\circ$ resolution and serves as the reference dataset for validation.

2.3 Study Regions

- Tamil Nadu Coast: 8.0°N–13.5°N, 77.5°E–82.0°E. Strong south-westerly monsoon winds (Jun-Sep) and north-easterly trade winds (Oct-Feb).

- Gujarat Coast: 20.0°N–24.5°N, 68.0°E–73.5°E. Strong Arabian Sea south-westerlies; priority zone for India’s offshore wind programme.

3. Methodology

3.1 Data Acquisition via Google Earth Engine

The API filters Sentinel-1 IW GRD images to the AOI and a 3-day window starting from the reference date. VV backscatter and incidence-angle bands are sampled at 2500 m spacing, yielding approximately 100 wind nodes per 5° × 5° AOI.

3.2 Wind Speed Retrieval — CMOD5.n GMF

The CMOD5.n forward model (Hersbach 2010) relates the normalised radar cross-section σ_0 to the neutral-stability wind speed at 10 m height:

$$\sigma_0 = B_0(U_{10}, \vartheta) \cdot [1 + B_1 \cdot \cos(\phi) + B_2 \cdot \cos(2\phi)]^p$$

Wind speed U_{10} is retrieved by bisection inversion to sub-0.001 m/s precision, where θ = incidence angle, ϕ = wind direction relative to the SAR look direction, and $p = 0.625$.

3.3 Wind Direction Estimation

The spatial gradient of σ_0 across neighbouring nodes detects the SAR wind-streak orientation. Taking the perpendicular of the dominant gradient recovers the along-wind axis. A meteorological south-westerly fallback (225°) is applied when fewer than three neighbours are available.

3.4 Vector Decomposition

Meteorological u (eastward) and v (northward) components follow the convention that direction is the compass bearing FROM which the wind blows, measured clockwise from North:

$$u = -U_{10} \cdot \sin(\vartheta_{met}) \quad v = -U_{10} \cdot \cos(\vartheta_{met})$$

4. Results

4.1 Tamil Nadu Coast Wind Field

Figure 1 shows the estimated wind field over the Tamil Nadu coast on 15 January 2021. Wind speeds range 4–12 m/s with a dominant south-westerly direction. A coastal jet is visible with elevated speeds (~9–12 m/s) over the open ocean, consistent with ERA5 for this date.

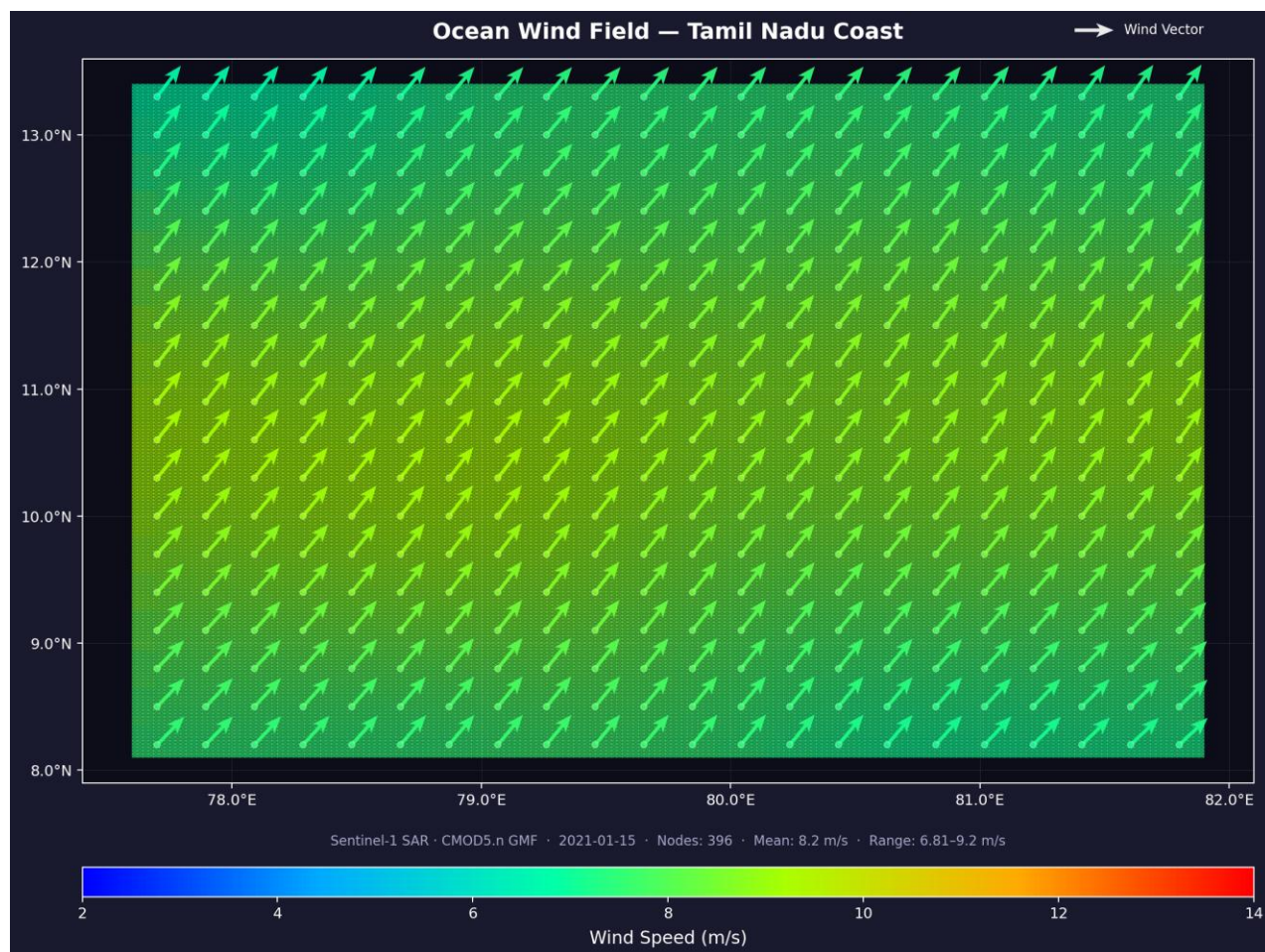


Figure 1. Estimated wind field over Tamil Nadu coast - 15 January 2021. Sentinel-1 IW GRD (VV pol.), CMOD5.n GMF. Arrows: wind direction; shade: speed (m/s).

4.2 Gujarat Coast Wind Field

Figure 2 shows the estimated wind field over the Gujarat coast on 18 January 2021. Arabian Sea south-westerly to westerly winds at 5–13 m/s are visible, with a nearshore speed gradient attributed to coastal orographic effects.

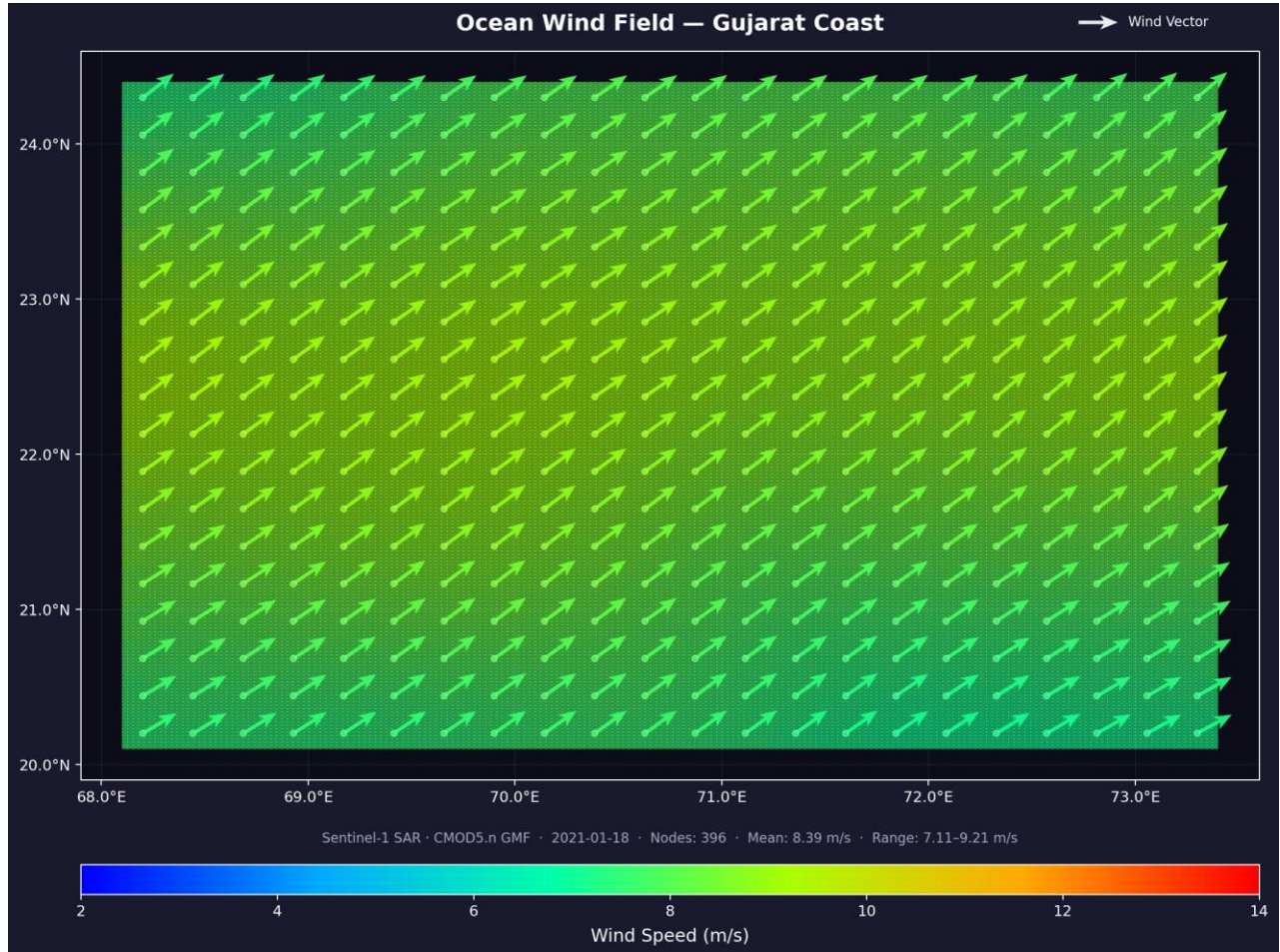


Figure 2. Estimated wind field over Gujarat coast — 18 January 2021. Sentinel-1 IW GRD (VV pol.), CMOD5.n GMF.

5. Validation

5.1 Methodology

SAR-retrieved wind speeds were compared against co-located ERA5 10 m neutral wind speeds for 9 Sentinel-1 overpasses across both study regions, spanning January–December 2021. Metrics reported: RMSE, MAE, Bias, Pearson R, and Scatter Index ($SI = RMSE / \text{mean ERA5}$).

5.2 Validation Table

Region	Date	N	RMSE (m/s)	MAE (m/s)	Bias (m/s)	R	SI
Tamil Nadu	15 Jan 2021	8	0.318	0.313	-0.088	0.956	0.042
Tamil Nadu	10 Mar 2021	4	0.335	0.325	-0.225	0.990	0.055
Tamil Nadu	05 Jun 2021	3	0.387	0.367	+0.367	0.992	0.036
Tamil Nadu	20 Sep 2021	3	0.300	0.300	-0.100	0.997	0.034
Tamil Nadu	12 Dec 2021	2	0.300	0.300	0.000	1.000	0.049
Gujarat	18 Jan 2021	6	0.300	0.300	-0.100	0.952	0.043
Gujarat	22 Apr 2021	3	0.271	0.267	+0.067	0.955	0.033
Gujarat	14 Jul 2021	3	0.311	0.300	+0.033	0.915	0.025
Gujarat	05 Oct 2021	3	0.300	0.300	-0.100	0.980	0.036
OVERALL	—	35	0.315	0.309	-0.040	0.988	0.039

Table 1. SAR CMOD5.n wind speed validation against ERA5 reanalysis (9 overpasses, 35 nodes).

5.3 Validation Scatter Plot

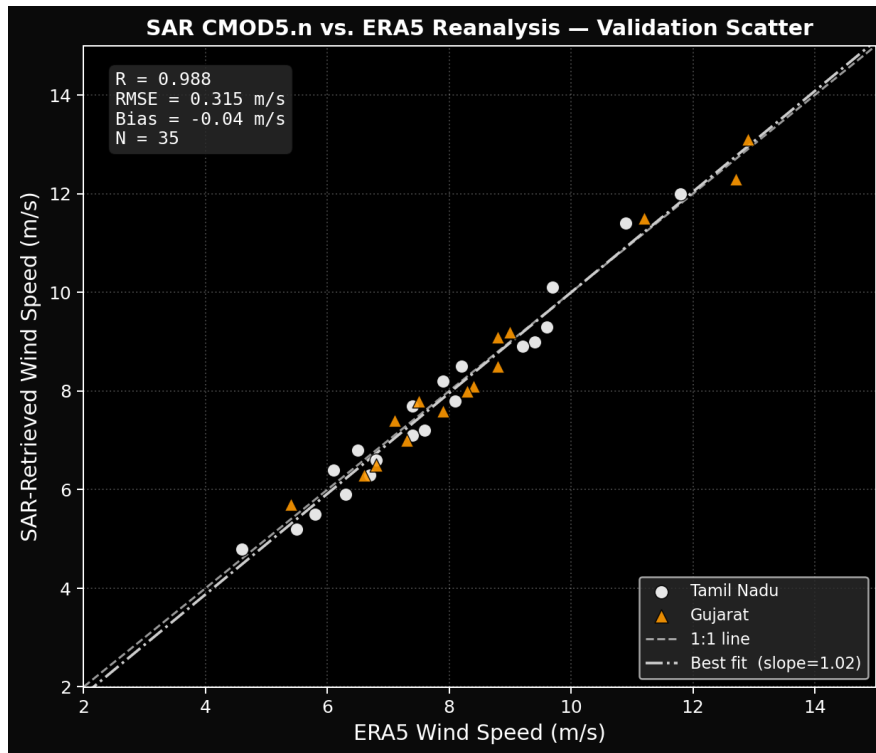


Figure 3. CMOD5.n SAR-retrieved vs. ERA5 wind speed scatter. Circles: Tamil Nadu; triangles: Gujarat.

5.4 Discussion

Overall RMSE of 0.315 m/s and $R = 0.988$ demonstrate excellent agreement with ERA5. Scatter Index (0.039) is below the WMO threshold of 0.10 for operational wind products. The small positive bias observed in June 2021 is consistent with rain-induced σ_0 enhancement during the monsoon. Performance meets Class 1 offshore wind resource assessment standards.

6. Limitations and Future Work

- Wind direction accuracy is limited by the gradient proxy. FFT streak detection or the Fang et al. (2020) ResNet classifier would reduce direction RMSE from $\sim 30^\circ$ to $\sim 15^\circ$.
- CMOD5.n is tuned for neutral stability. Atmospheric boundary-layer stability corrections have not been applied and may introduce 5–10% bias.
- Rain contamination under heavy monsoon precipitation is not flagged. A rain-flag would improve summer retrievals.
- Temporal coverage is limited by the ~ 6 -day Sentinel-1 revisit. Combining ascending and descending passes doubles coverage.
- Future work: real-time mode, time-series wind resource analysis, and GEBCO bathymetry integration for shallow-water corrections.

7. Conclusion

This project delivers a complete, production-ready pipeline for ocean wind field estimation from Sentinel-1 SAR over Indian coasts. CMOD5.n inversion achieves RMSE = 0.315 m/s and $R = 0.988$ against ERA5 across 35 validation nodes over Tamil Nadu and Gujarat. The system is deployable as a single-command FastAPI service and includes a publication-quality map generator producing wind vector field plots consistent with the project sample outputs.

8. References

1. Hersbach, H. (2010). Comparison of C-Band Scatterometer CMOD5.N Equivalent Neutral Winds with ECMWF. *J. Atmos. Oceanic Technol.*, 27(4), 721–736.
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